INCH-POUND
MIL-DTL-49285/21B
27 March 2018
SUPERSEDING
MIL DTL 49285/21A
1 March 2010

DETAIL SPECIFICATION SHEET

CABLE, SPECIAL PURPOSE, ELECTRICAL, THREE INDIVIDUALLY SHIELDED CONDUCTORS OF 22 (7 X 30) AWG

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

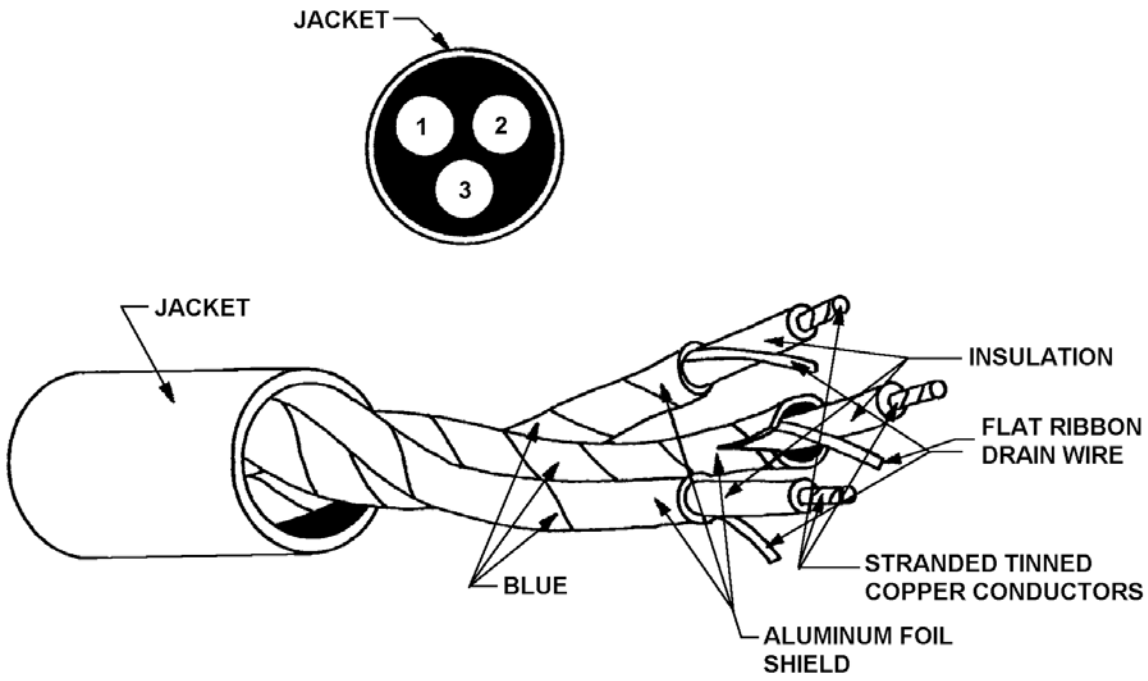


FIGURE 1. Cable illustration.

Engineering information: See table I.

Note: Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.

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TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductor and shield	55 pF/ft (picofarads/foot)
Continuous working voltage	350 V rms maximum between conductors
Nominal capacitance between adjacent shields	150 pF/ft
Continuous working voltage between adjacent shields	50 volts maximum
Shield dc resistance	40 ohms/1,000 feet maximum
Drain wire resistance	102.0 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of conductors	Three individually shielded
Conductor type and wire size	Stranded tinned copper 22 AWG
Conductor stranding	7 X 30 AWG
Conductor insulation	Polypropylene
Conductor insulation thickness	0.014 inch nominal thickness
Drain wire type and size	Tinned cadmium bronze tinsel
Drain wire thickness	.003 inch minimum .004 inch maximum
Drain wire width	.025 inch minimum .031 inch maximum
Jacket material	Polyvinyl chloride
Jacket thickness	0.030 inch
Finished cable diameter	0.215 inch maximum outside diameter
Cable style (UL)	Not applicable
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay lengths	2.0 twists per foot ± 10 percent

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REQUIREMENTS:

Design and construction: (see table II):

Shield color code. The shield color shall be blue.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Conductor color code.

1st conductor: Black
2nd conductor: Red
3rd conductor: Clear

Flammability. The cable shall pass UL 2556 flame test requirement.

Shield integrity test.* Each shielded conductor is to be tested as if it were a pair.

Frequency	Response
100 kHz	115 dB or more below reference
500 kHz	100 dB or more below reference
1 MHz	94 dB or more below reference
5 MHz	76 dB or more below reference
10 MHz	65 dB or more below reference
15 MHz	59 dB or more below reference
20 MHz	55 dB or more below reference
25 MHz	50 dB or more below reference
30 MHz	47 dB or more below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.

At 100 MHz: At least 34 dB below reference.

Crosstalk test limits. Crosstalk testing is not required for this configuration.

Insulation resistance between shields: 100 megohms-1,000 feet (minimum).

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 85 ± 5.

Part or Identifying Number (PIN): The PIN shall be M49285/21.

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Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-49285, this document references the following: UL 2556.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA – CC

Preparing activity:
DLA - CC

Review activities:
Army – MI
Air Force – 71
DLA - CQ

(Project 6145-2018-029)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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